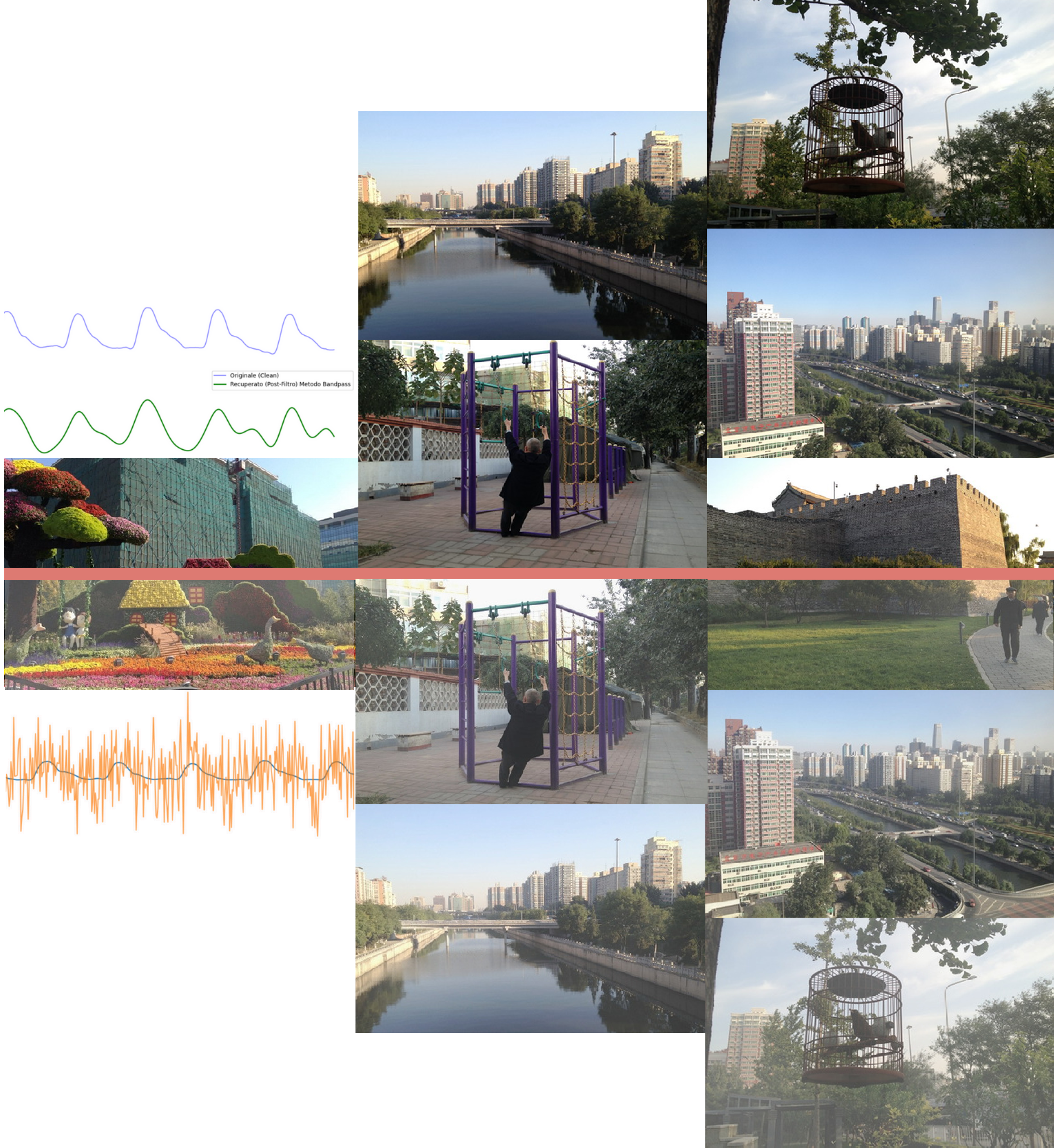
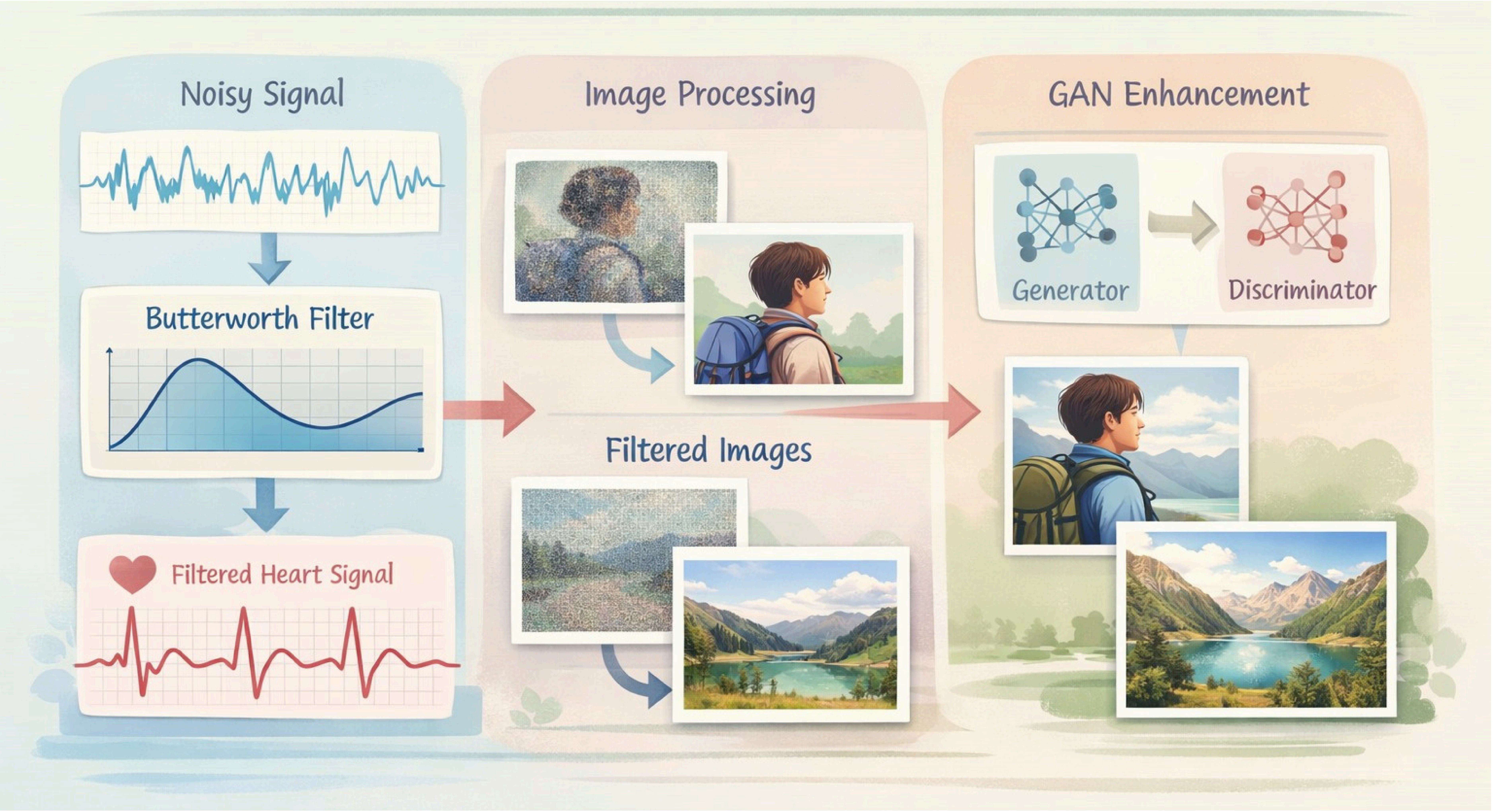
RECOVERY AND INFORMATION EXTRACTION FROM DEGRADED SOURCES

CRISTINA PAPI
MARCO ROSATO




Noisy Signal
Butterworth Filter
Filtered Heart Signal
Image Processing
Filtered Images
GAN Enhancement
Generator
Discriminator

1 Module

HEART RATE EXTRACTION FROM A PPG SIGNAL

Dataset: BIDMC PPG and Respiration (physio net)
Signals frequency: 125 Hz
Numerics frequency: 1 Hz

Analysis

Heart rate extraction from a PPG signal is performed through a denoising stage based on a Butterworth filter, followed by frequency-domain analysis using FFT and peak detection to compute the BPM:

WHITE GAUSSIAN NOISE → BUTTERWORTH FILTER → HR ESTIMATION → VALIDATION → RESULTS

In order to make correct analysis, we divided data into different windows.
In fact, in real-world scenarios, the heart rate is dynamic, accelerating and decelerating due to physiological factors such as arrhythmias, respiration, and stress. **Segmentation** allows us to track these heart rate variations over time rather than just calculating a single static average.

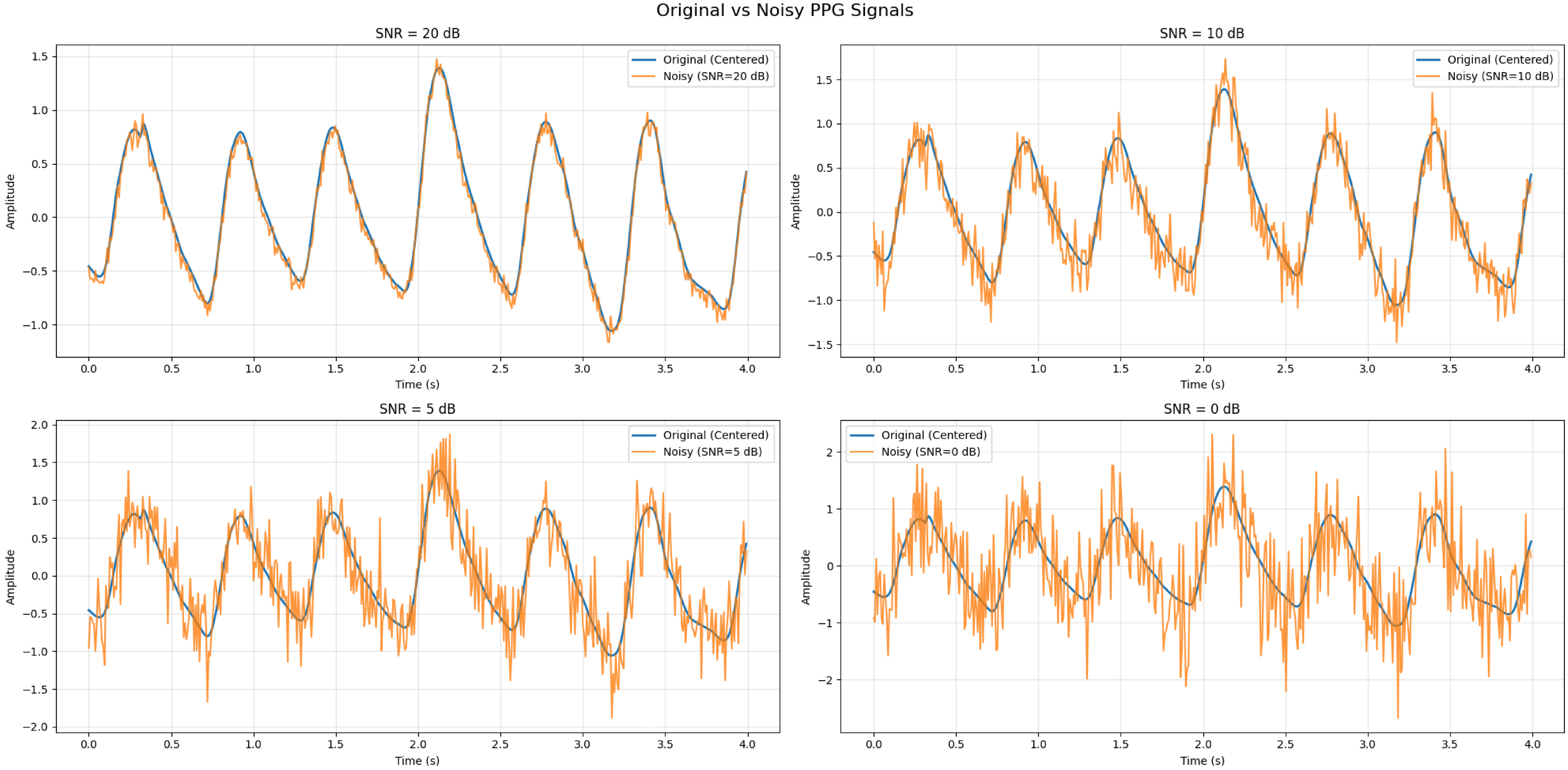
The use of an overlap (STEP) is useful to prevent the loss of information

White Gaussian Noise

We chose White Gaussian Noise because it is a standard statistical model in signal processing. It allows us to simulate random noise that affects all frequencies equally, mimicking the thermal and electronic noise found in real-world PPG sensors

We added synthetic noise to arrive to these 4 level of SNR (metric used to compare the level of a desired signal to the level of background noise):

- 20 dB: high quality signal
- 10 dB: moderate noise
- 5 dB: significant noise
- 0 dB: extreme noise (bad)

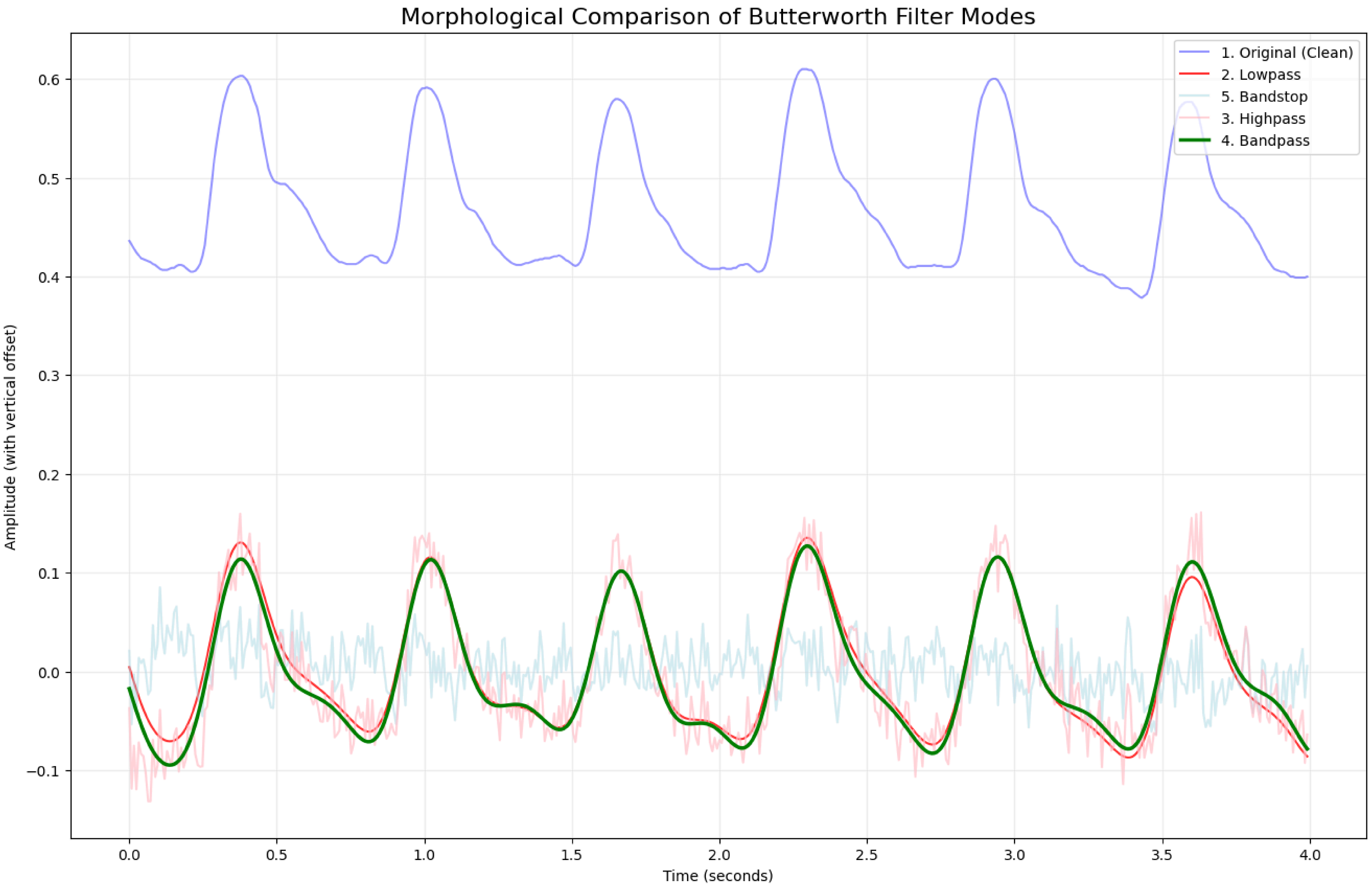

Original vs Noisy PPG Signals
SNR = 20 dB
SNR = 10 dB
SNR = 5 dB
SNR = 0 dB
Original (Centered)
Noisy (SNR=20 dB)
Noisy (SNR=10 dB)
Noisy (SNR=5 dB)
Noisy (SNR=0 dB)
Amplitude
Time (s)

Butterworth filter

The Butterworth filter is selected because it offers a very smooth response in the passband, ensuring that noise is suppressed without distorting the physiological shape of the PPG pulses (it preserves the integrity and morphology of the heartbeats while effectively cutting out unwanted interference).

4 types of filter

- LOW-PASS
- BAND-PASS (0.5 - 4 HZ)
- HIGH-PASS
- BAND-STOP



Morphological Comparison of Butterworth Filter Modes
1. Original (Clean)
2. Lowpass
5. Bandstop
3. Highpass
4. Bandpass
Amplitude (with vertical offset)
Time (seconds)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Heart Rate Estimation Algorithms

The **Fast Fourier Transform (FFT)** is a mathematical algorithm that converts the PPG signal from its original Time Domain (amplitude over time) to the Frequency Domain (power distribution across frequencies). It Identifies the dominant frequency peak**.** It separates the primary cardiac component from low-frequency baseline wander and high-frequency sensor noise and considers the entire window.

By identifying the frequency with the highest magnitude within the physiological range (typically 0.5 to 4.0 Hz, corresponding to 30–240 BPM), we can estimate the heart rate using the formula

BPM = Peak Frequency (Hz)× 60

The **Peaks Detection** is a time-domain technique used to identify the specific moments when the heart ejects blood into the peripheral vessels, known as systolic peaks. In a PPG signal, these are the local maxima of the waveform. It tracks the actual shape of the pulse wave. It is highly sensitive to noise.

The algorithm identifies the time indices of each valid systolic peak. It then calculates the Inter-Beat Interval (IBI), which is the time elapsed between two consecutive peaks. The heart rate is derived from the average of these intervals:

BPM = 60 / Mean IBI (seconds)

Metrics

MAE

It quantifies the average magnitude of the difference between the estimated BPM (calculated by our algorithms) and the ground truth BPM (provided by the dataset)

Formula:

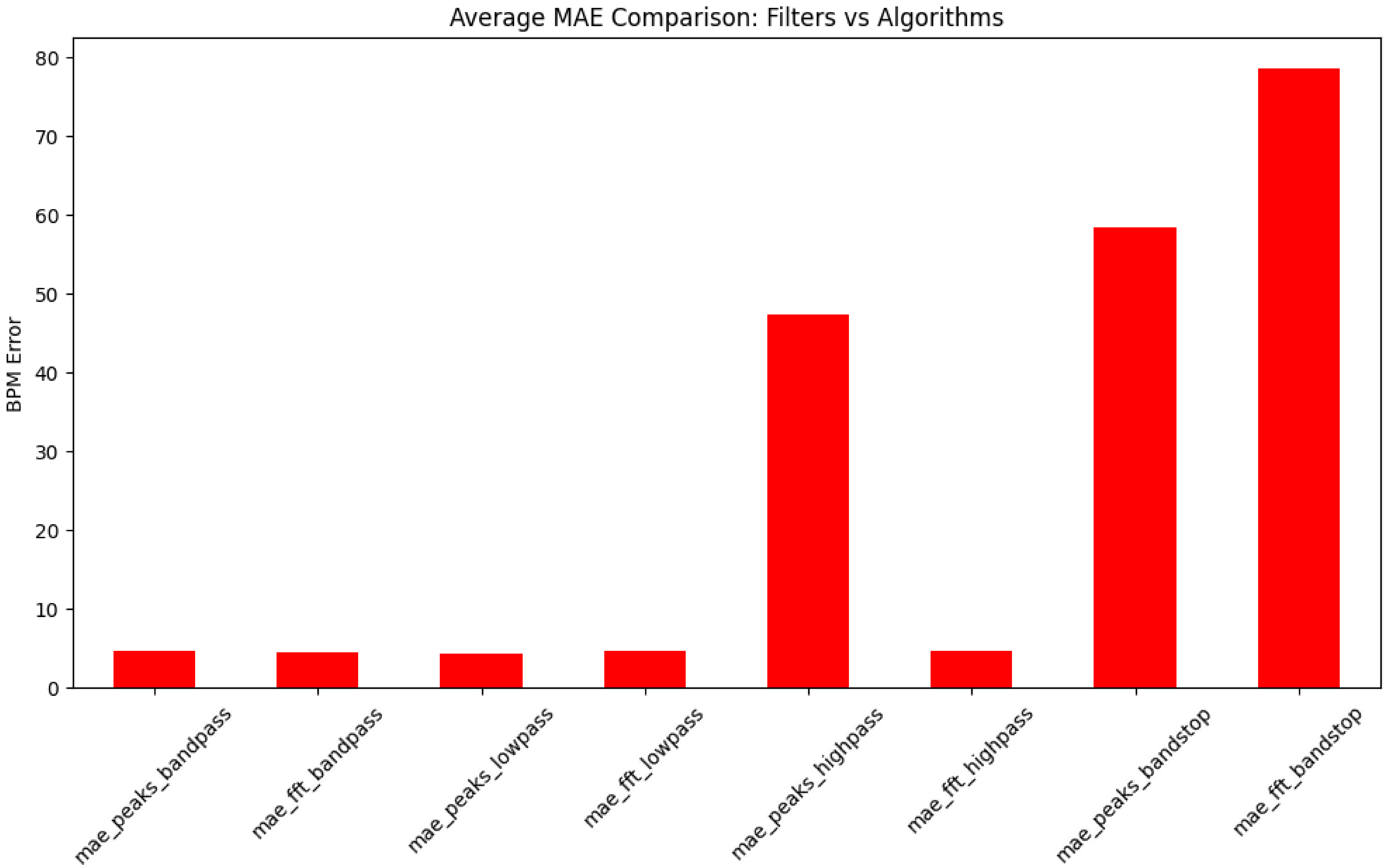
$$MAE = \frac{1}{n}\sum_{i=1}^{n} |BPM_{calc,i} - BPM_{true,i}|$$

SNR

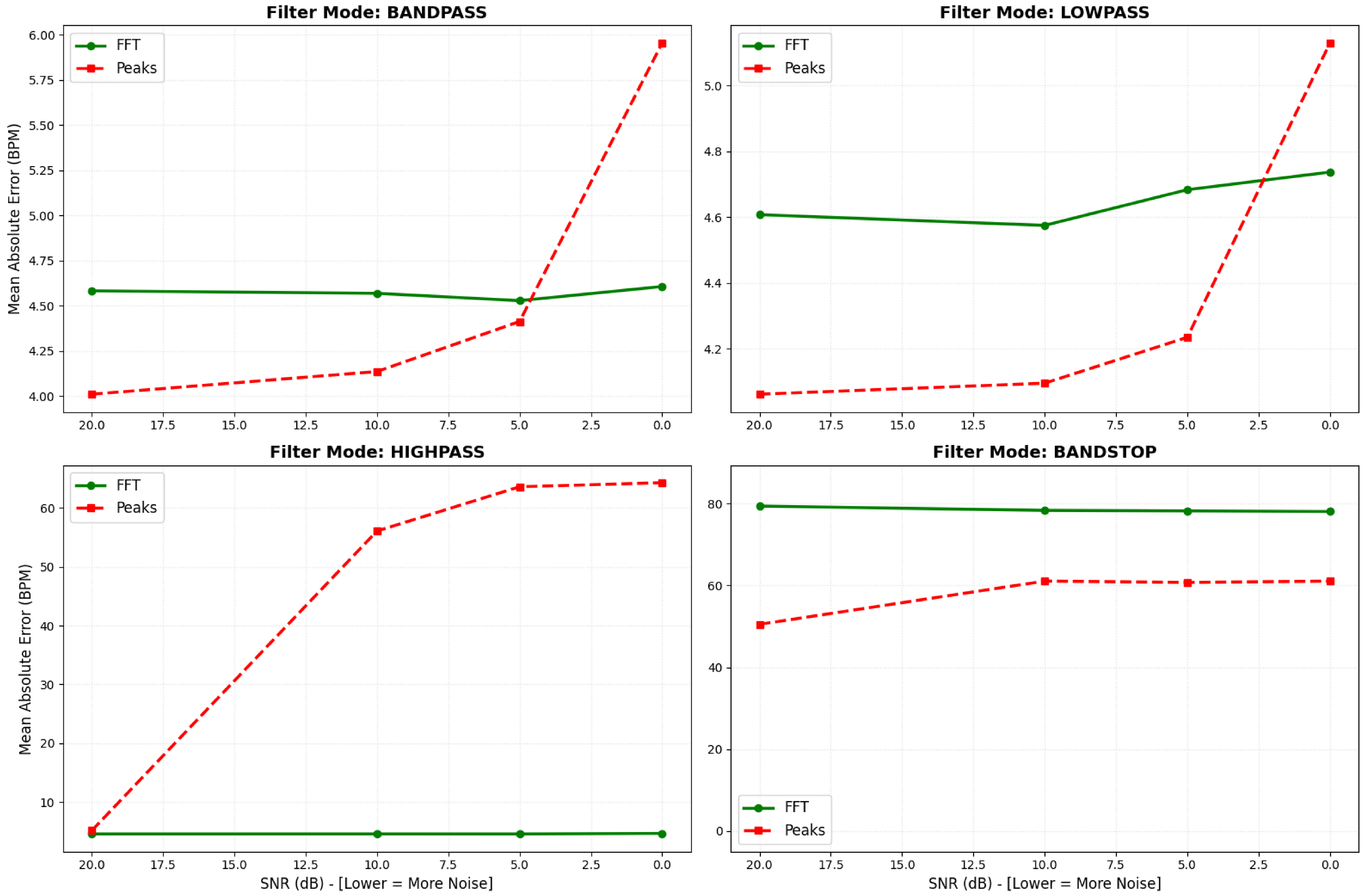
It is a standard metric used in science and engineering to compare the level of a desired signal to the level of background noise. It is defined as the ratio of signal power to noise power, often expressed in decibels (dB).
If positive -> signal power is higher than the noise power (good quality)

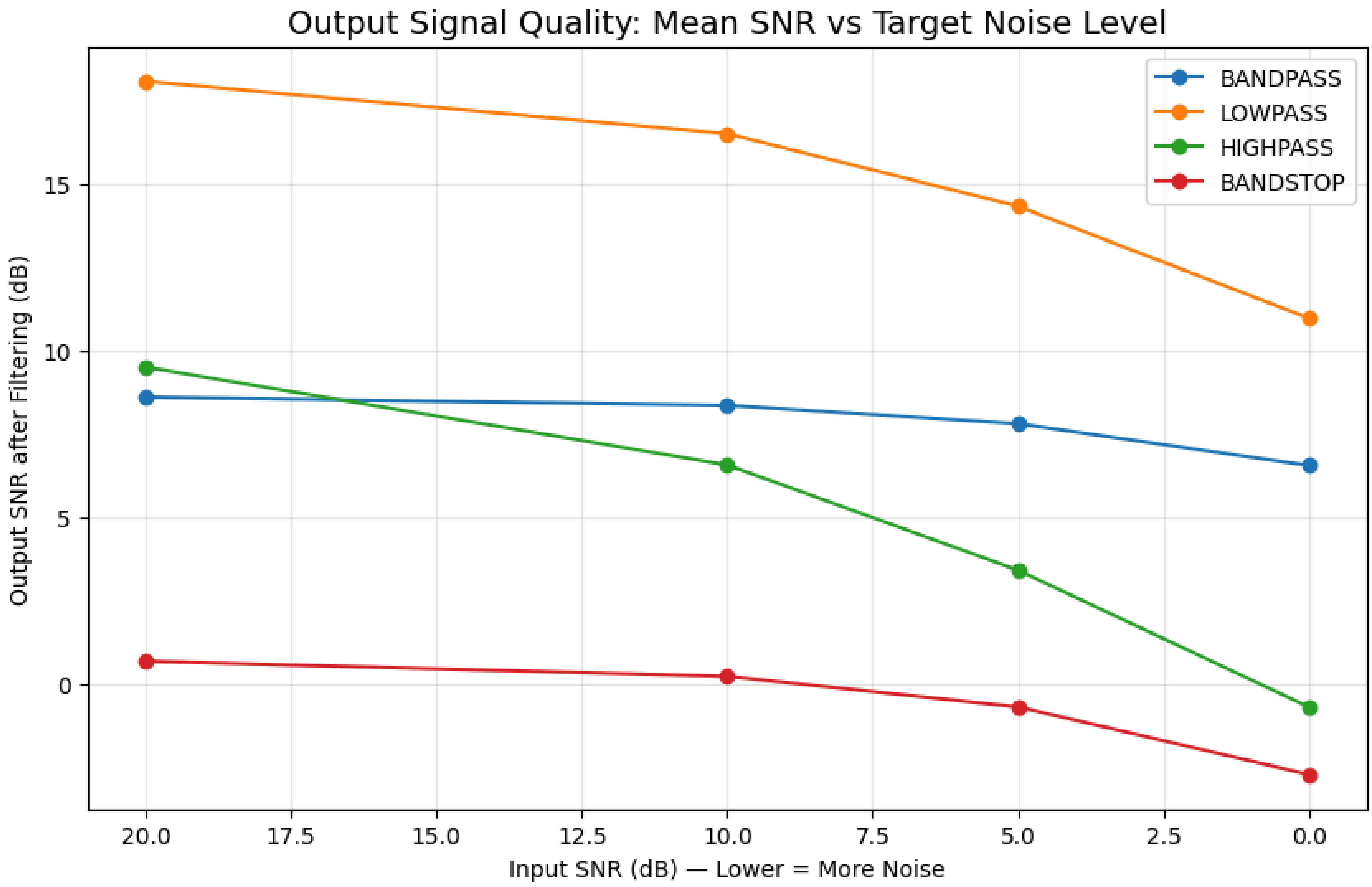
Formula:

$$SNR_{dB} = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$


Average MAE Comparison: Filters vs Algorithms
BPM Error
0
10
20
30
40
50
60
70
80
mae_peaks_bandpass
mae_fft_bandpass
mae_peaks_lowpass
mae_fft_lowpass
mae_peaks_highpass
mae_fft_highpass
mae_peaks_bandstop
mae_fft_bandstop

Robustness Analysis: MAE vs SNR for different Filter Modes




Output Signal Quality: Mean SNR vs Target Noise Level
BANDPASS
LOWPASS
HIGHPASS
BANDSTOP
Output SNR after Filtering (dB)
15
10
5
0
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Input SNR (dB) — Lower = More Noise

2 Module

IMAGE DEHAZING
DARK CHANNEL PRIOR

Atmosferic Scattering Model

Image degradation in foggy scenes can be described by the atmospheric scattering model:

I(x) = J(x)t(x) + A(1-t(x))

Where:

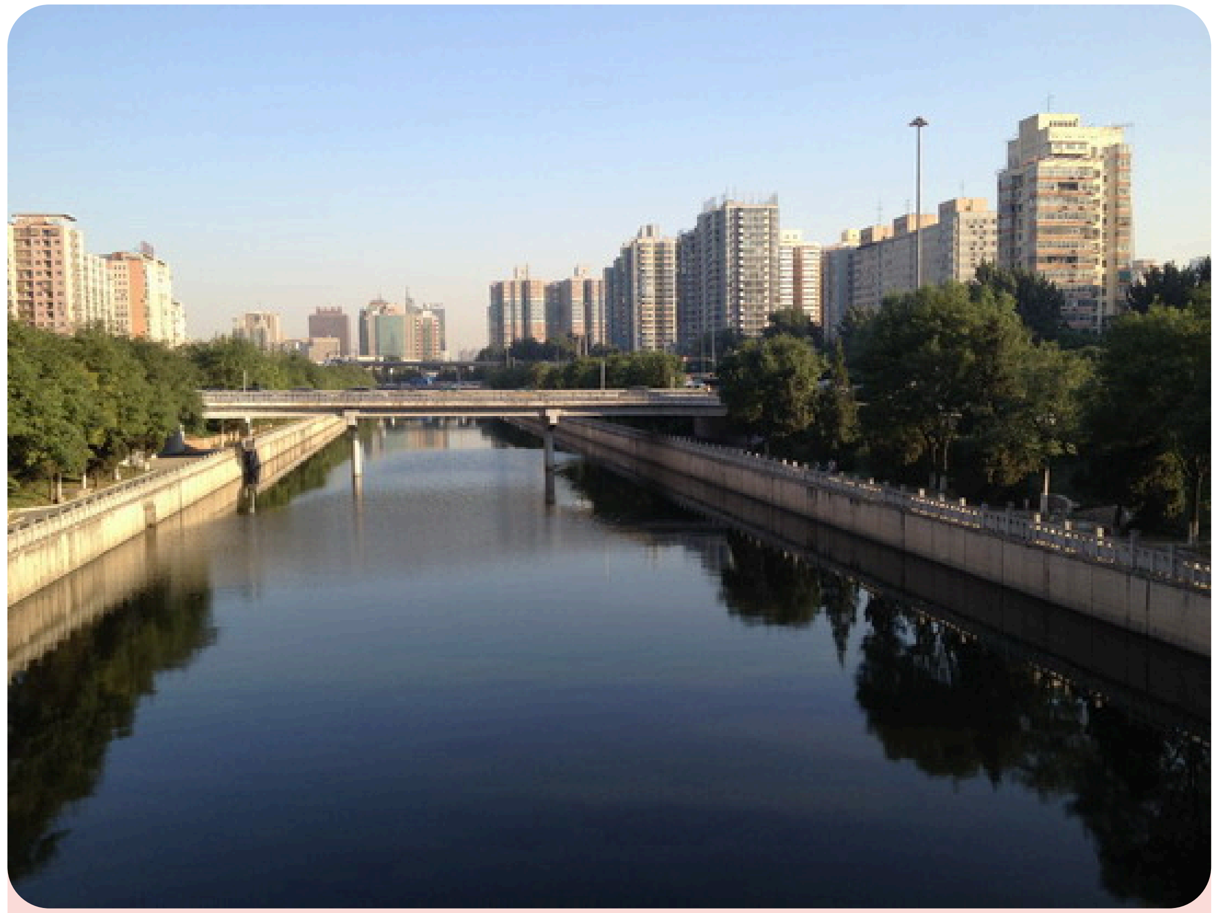
- **I(x)** is the **observed hazy image**;
- **J(x)** is the **unknown clear image**;
- **t(x)** ∈ [0,1] is the transmission map;

$t(x) = e^{-\beta d(x)}$. A formulation based on the Beer-Lambert's Law.

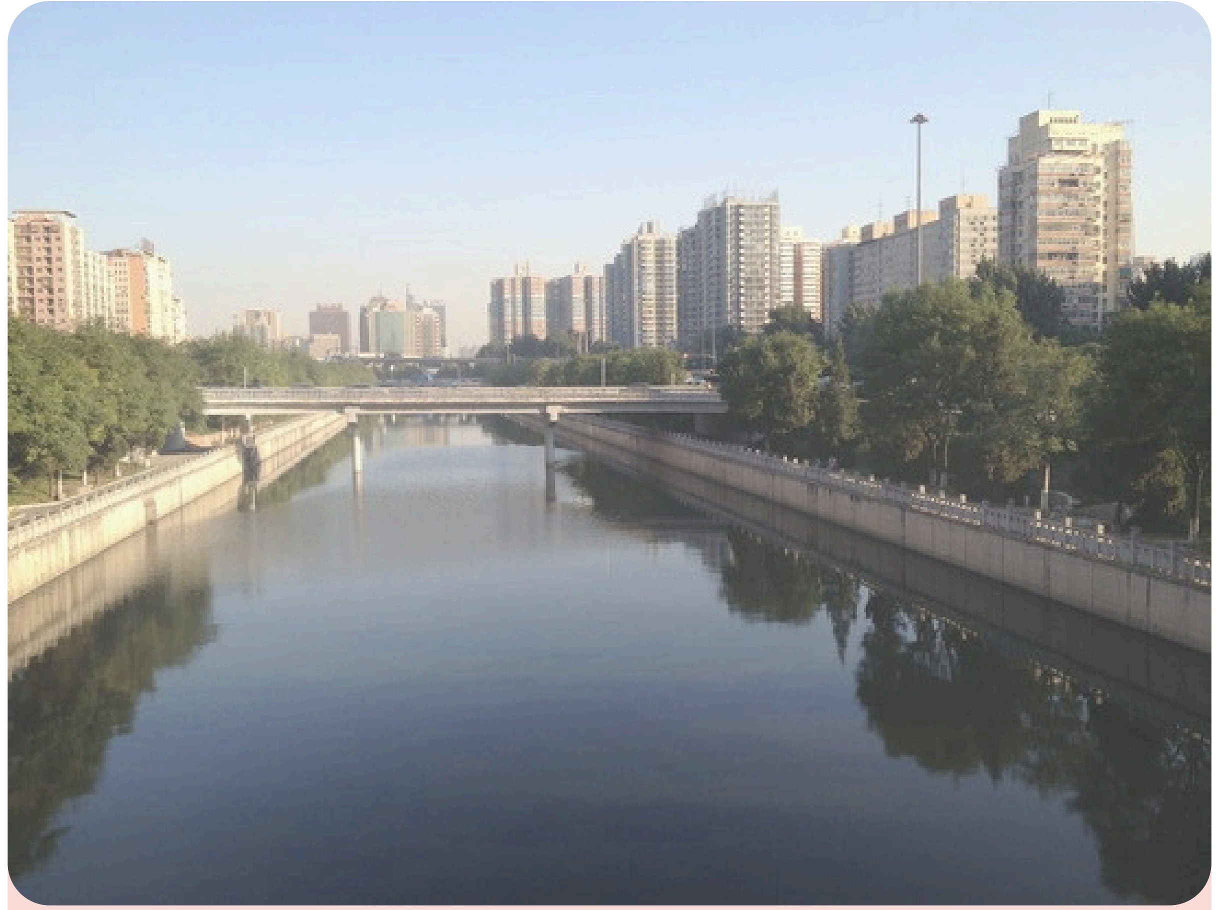
Recovering (J(x)) from (I(x)) is an underdetermined problem, as both (J(x)) and (t(x)) are unknown.

Dataset

Synthetic Objective Testing Set (SOTS) [RESIDE]. In particular, we used the OutDoor images (492 pairs of images).



CLEAR VERSION



HAZE VERSION

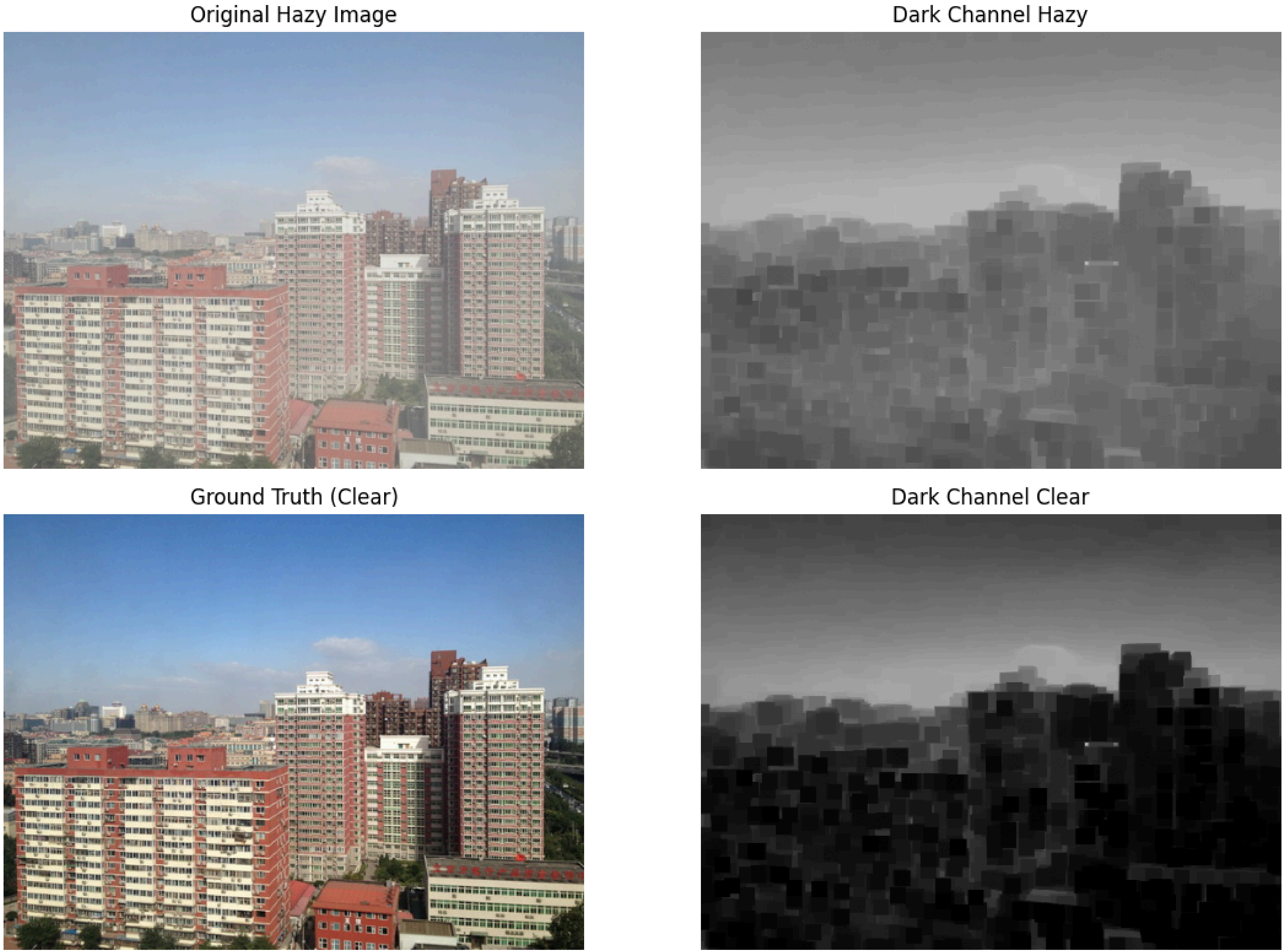
Dark Channel Prior (DCP)

DCP (He et al. (2009)) is based on the observation that in clear images, at least one color channel has low intensity in some pixels.

Min_channel(x) = min(R(x), G(x), B(x))

To provide robustness to noise and texture, it is implemented with patches (window) not single pixels.

DarkChannel(x) = min (neighboorhood (x))


Original Hazy Image
Dark Channel Hazy
Ground Truth (Clear)
Dark Channel Clear

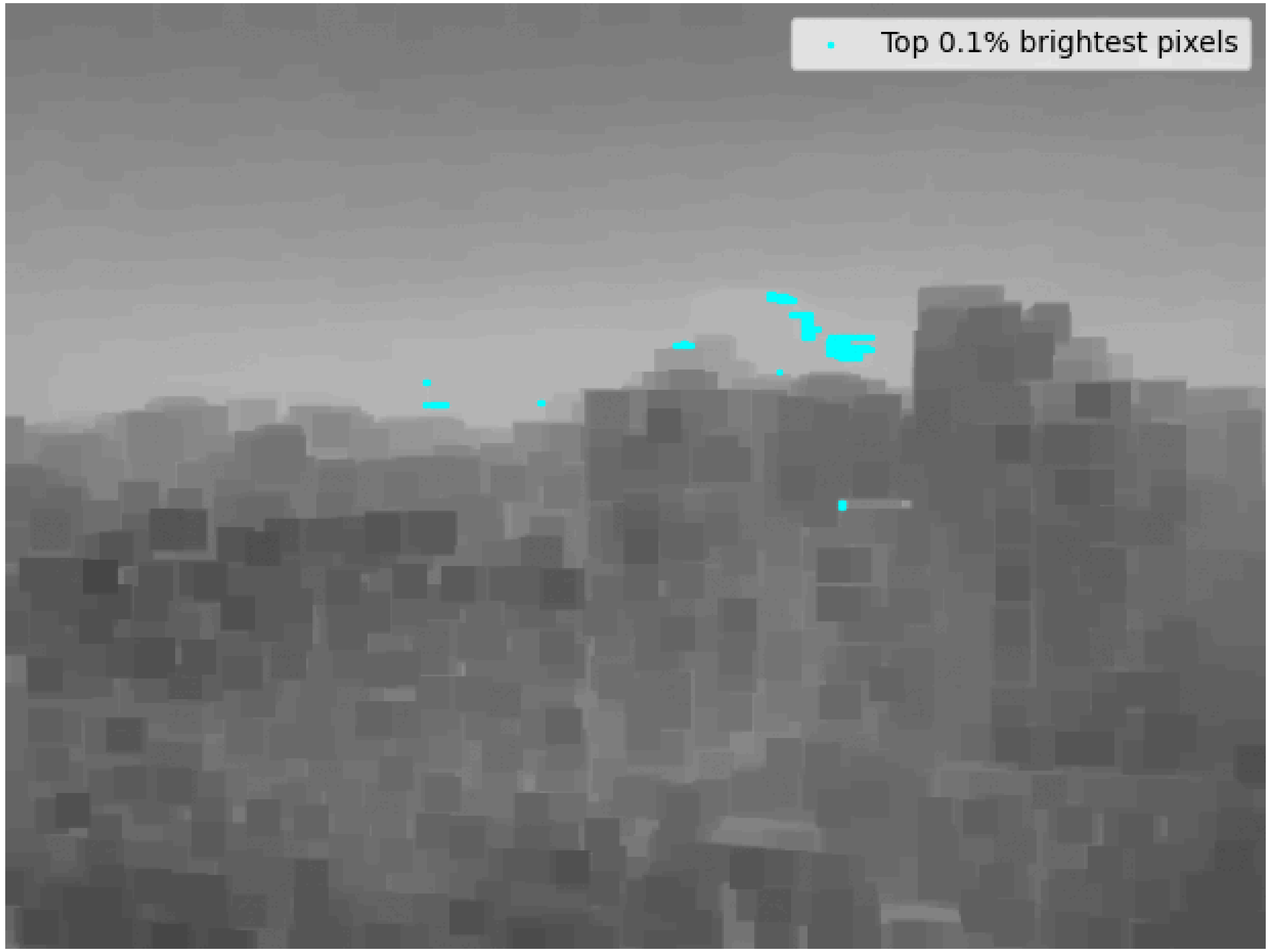
Atmospheric Light (A)

A represents the **intensity and color of the ambient illumination in the hazy scene**.

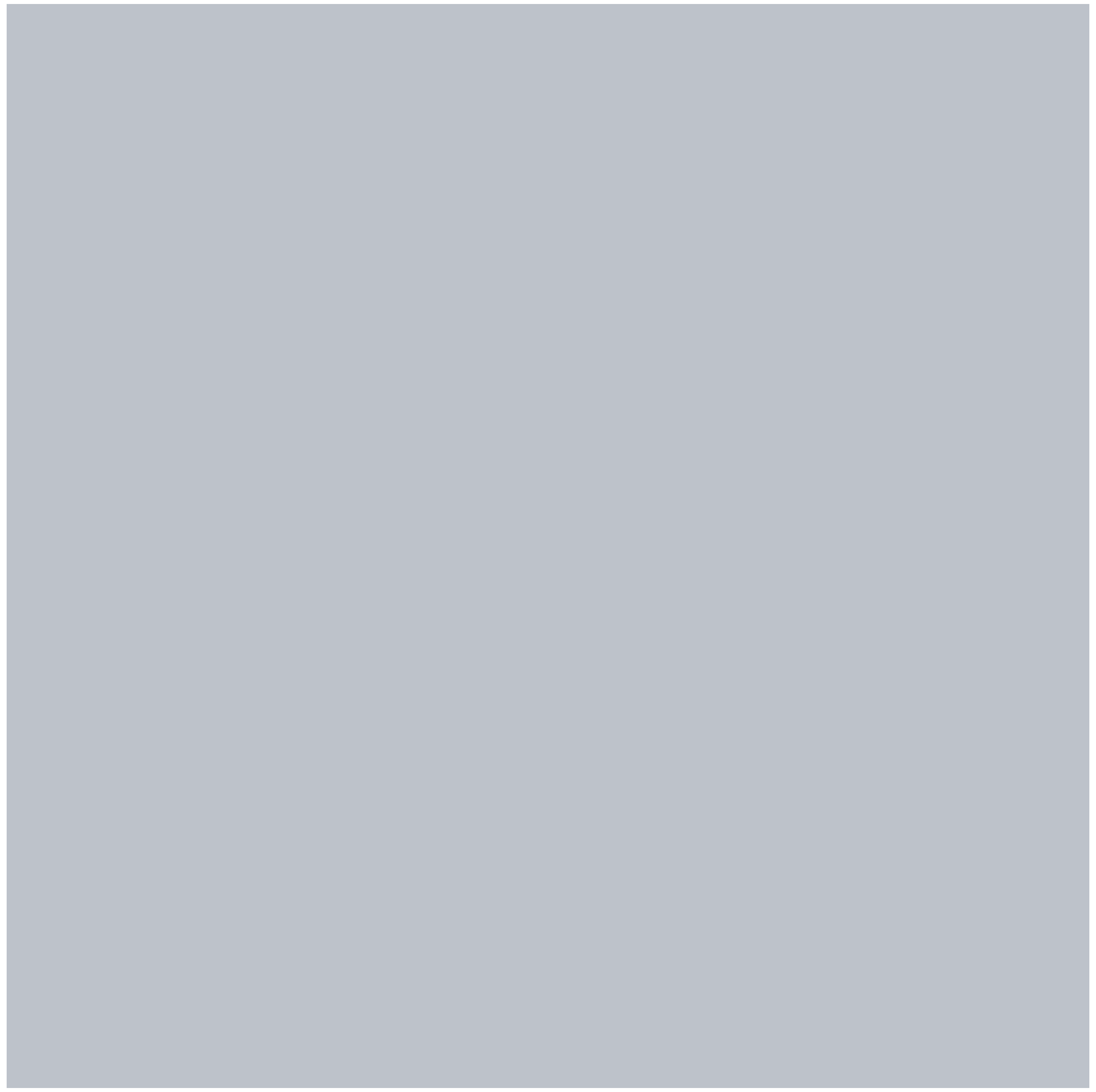
It is done in **two steps**:

- Identify the top 0.1% brightest pixels in the Dark Channel.
- Median value of this subset in the original RGB image.

Top 0.1% brightest pixels in the dark channel (227 pixels)



Color of the Haze (A)
RGB: [189 194 202]



Transmission Map t(x)

The transmission map tells us how much of the original light from the objects actually “survies” the journey through the haze to reach the camera.

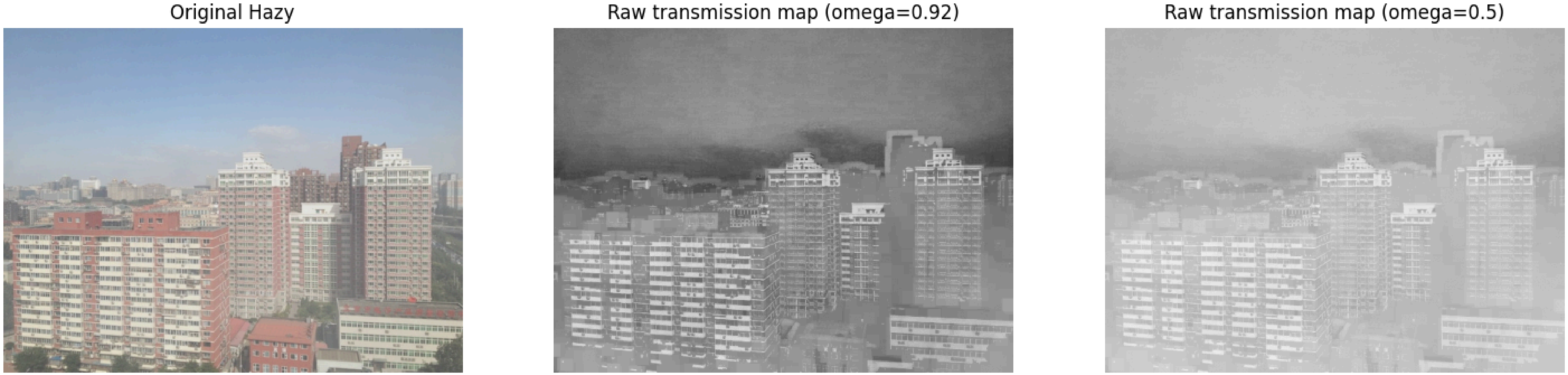
- t(x) = 1 → No haze, full visibility
- t(x) = 0 → full haze, no visibility

It is estimated as:

$$t(x) = 1 - \omega \cdot DarkChannel(\frac{I(x)}{A})$$

ω is the parameter that prevents to remove 100% of the haze

We also Implemented an adaptive constraint using a transition luminance mask to prevent artifacts in the sky while preserving the recovery in the rest of the scene.

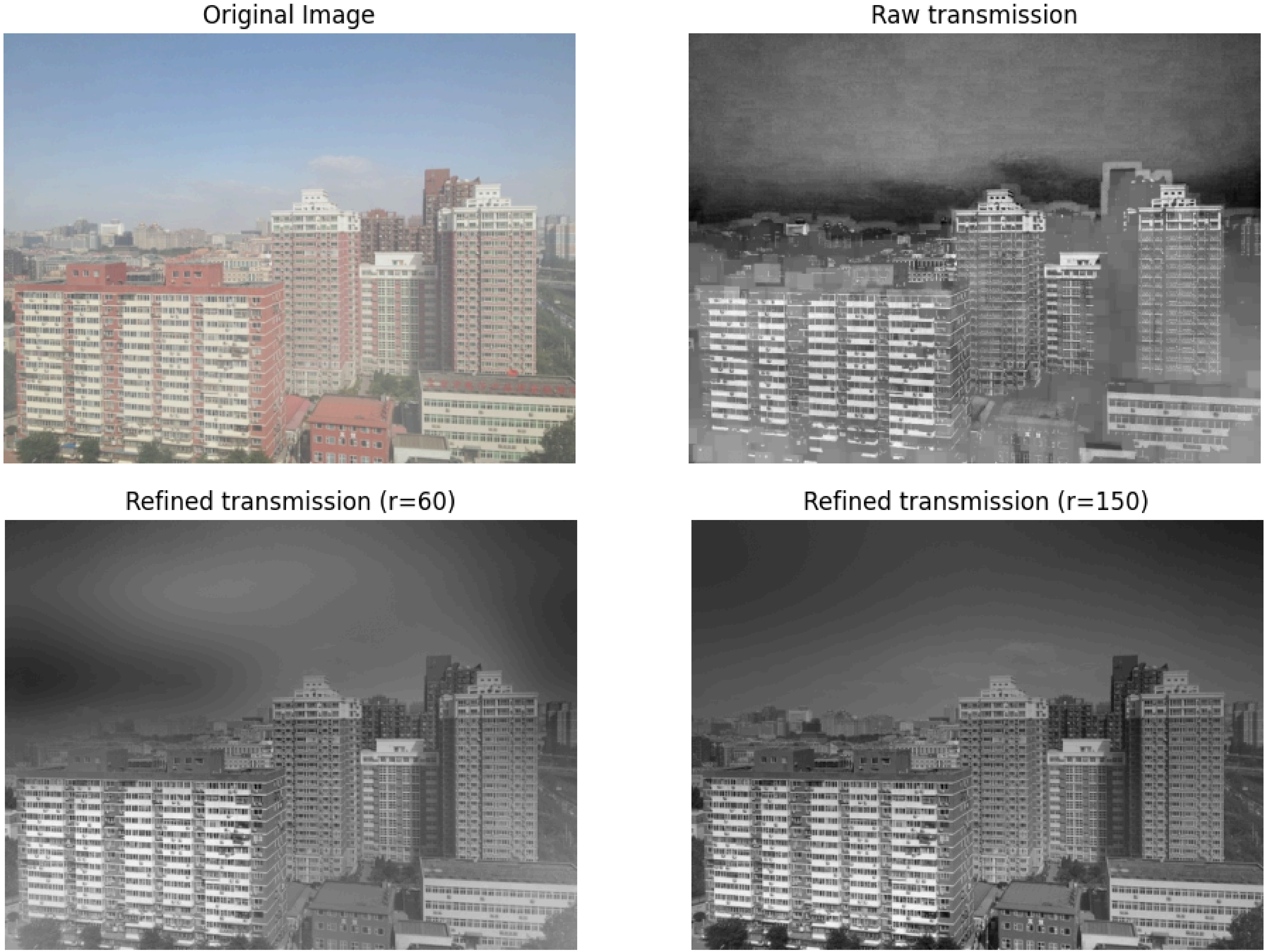


Guided Filter

If the scene recovery was done on the previous transmission map, there would be artifacts and halo effects. To prevent this, we used a **Guided Filter** that will:

- **Smooths homogeneous regions.**
- **Preserves sharp edges**

We implemented this filter with “cv2.ximgproc.guidedFilter” from OpenCV


Original Image
Raw transmission
Refined transmission (r=60)
Refined transmission (r=150)

Scene Recovery

Now we have all the “ingredients”! From the original formula, we can obtain the **original image J(x)**:

$$J(x) = \frac{I(x) - A}{max(t(x), t_0)} + A$$

At denominator we introduce a lower bond t0 to avoid numeric instability and the amplification of the noise in low visibility part of the image.

From our result, we decided also to **add a bit of Gamma and Saturation** to increase the final quality results.

Metrics:

AVERAGE	PSNR (db)	SSIM
Hazy (input)	15.97	0.8151
DCP Recovered	22.95	0.9266
DCP + Enhancement (gamma & saturation)	**23.81**	**0.9266**

Visualization

Input | Dehazed (raw) | Dehazed (G+S) | GT

3 Module

SUPER-RESOLUTION & RESTORATION SRRGAN

Super-Resolution & Image Restoration

The task is to reconstruct a high-resolution (HR) image from a low-resolution (LR). This is an img2img task.

HR full

HR Patch

LR Patch (degraded)

Dataset

We used 2 different dataset for this task:

- Train & Validation: DF2K (3,450 training images, 100 validation images)
- Test: BSDS100 (100 images)

IMAGE FROM TRAINING

IMAGE FROM TEST

Degratation pipeline

We did a blind restoration task. We applied randomic degration to the training and fixed degradation to the evaluation.

TRAINING

Pre-Degradation:
- Blur: Gaussian or Box
- Downsample (x2): bilinear, bicubic, area

Post-Degradation (random order):
- Noise: Gaussian or Poisson
- Jpeg (80% chance)

VALIDATION

Pre-Degradation:
- Blur: Gaussian
- Downsample (x2): area

Post-Degradation (fixed order):
- Noise: Gaussian
- Jpeg (always)

Model Architectures

Generator: U-NET with a “long” bottleneck. Takes a 128x128 patch and upsamples it to a 256x256 output.
Other features:

- Squeeze-and-Excitation
- Spatial Attention
- PixelShuffle
- Global Residual

Discriminator: PatchGAN with “SpectralNormalization”.
Maximum number of filters: 512 (32x32x512).
There is NO activation function!

Loss functions

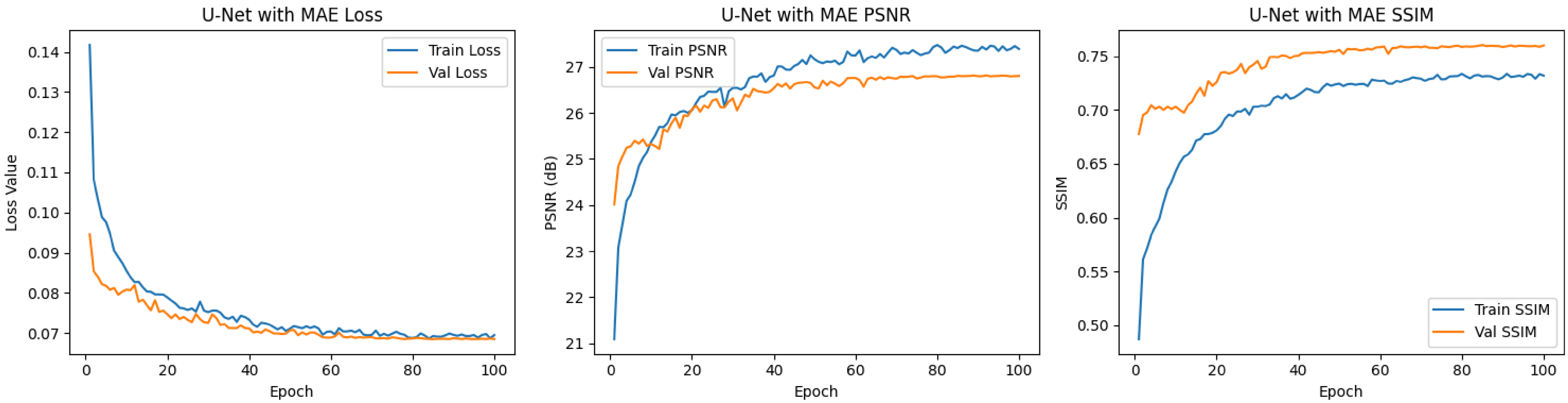
We used **three loss functions**:

MAE: PIXEL-WISE LOSS OF THE ABSOLUTE DIFFERENCE BETWEEN THE PREDICTED AND THE GT IMAGE.

PERCEPTUAL: USE A PRE-TRAINED VGG19 TO EXTRACT FEATURE FROM DIFFERENT LAYERS FROM BOTH THE PREDICTED AND GT IMAGE. MINIMIZE THE DIFFERENCES WITH MAE.

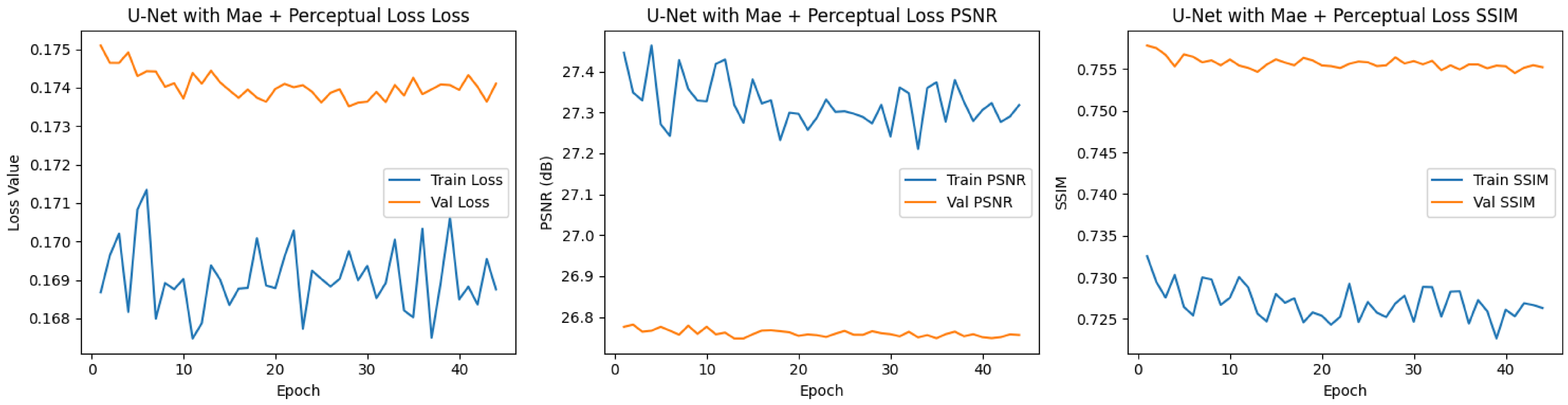
LEAST SQUARES GAN (LSGAN): TREATS THE DISCRIMINATOR AS A REGRESSION TASK, USING MEAN SQUARED ERROR (MSE) TO MINIMIZE THE DISTANCE BETWEEN PREDICTIONS AND BINARY TARGETS (1 FOR REAL, 0 FOR FAKE)

Pre-training: MAE



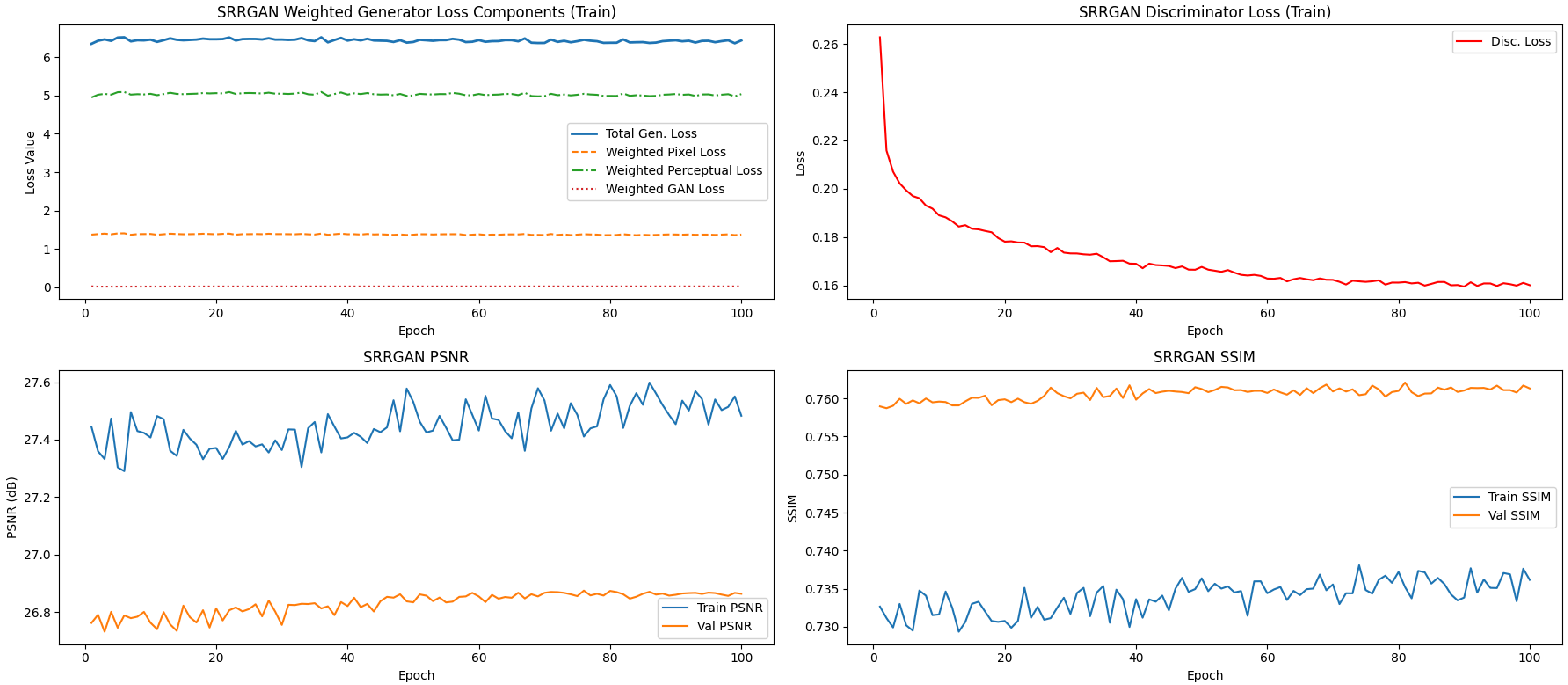
Perceptual Refinement (MAE + Perceptual)

Weight applied: $1 \cdot \mathrm{MAE}_{\mathrm{loss}} \quad , \quad 1 \times 10^{-3} \cdot \mathrm{Perceptual}_{\mathrm{loss}}$



Adversal Fine-tuning (MAE + Perceptual + GAN)

Weight applied: $20 \cdot \mathrm{MAE}_{\mathrm{loss}}$, $5 \times 10^{-2} \cdot \mathrm{Perceptual}_{\mathrm{loss}}$, $5 \times 10^{-2} \cdot \mathrm{GAN}_{\mathrm{loss}}$



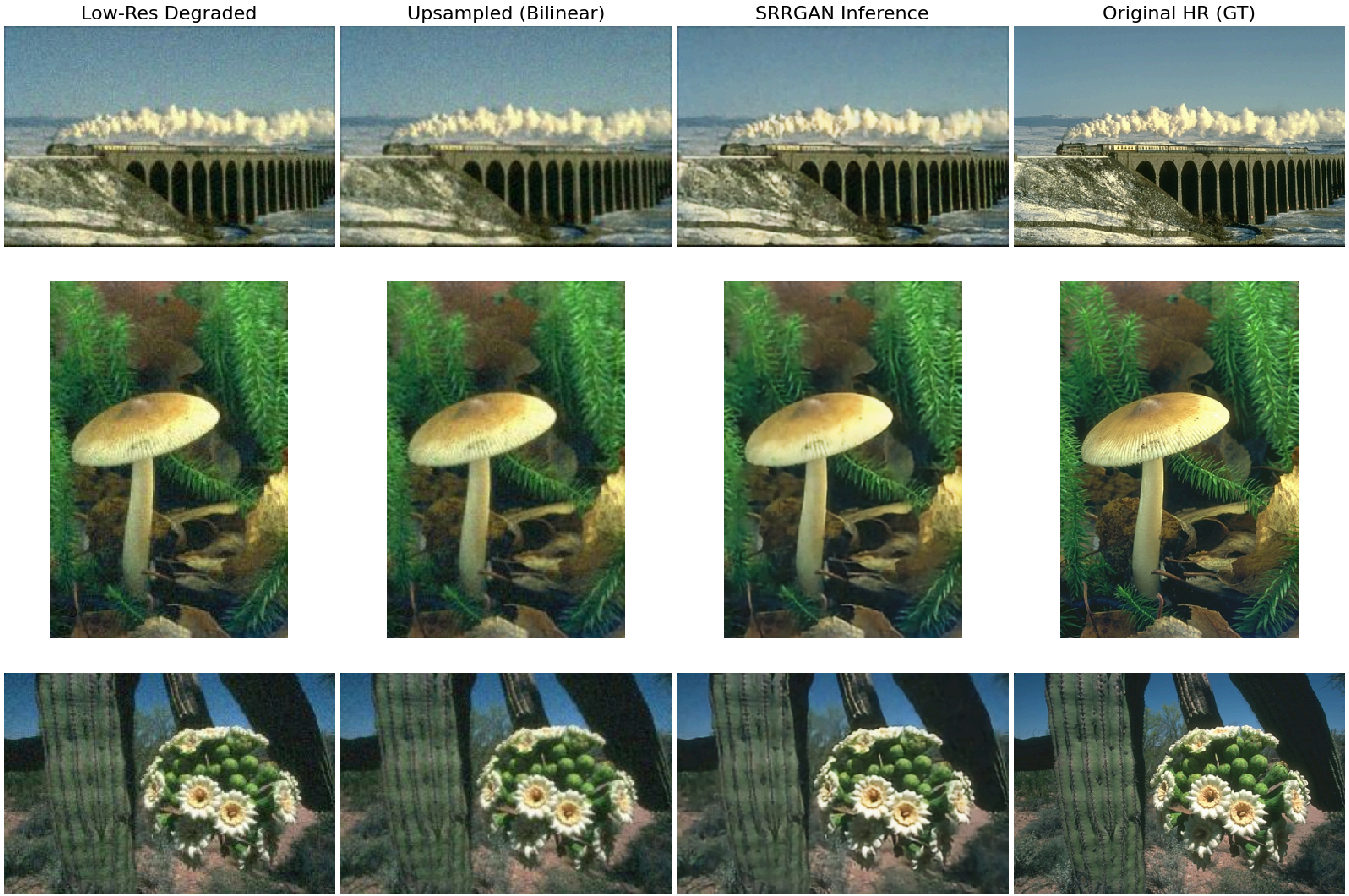
Evaluation Metrics:

We found the BEST GAN “checkpoint” at epoch 40 with a LPIPS = 0.3675.
The worst one is from epoch 4 with LPIPS = 0.3710.

Metric	**U-Net (MAE)**	**U-Net (MAE + Perceptual)**	**SRRGAN**
PSNR	25.6840	25.6607	**25.7109**
SSIM	0.7152	0.7130	**0.7171**
MAE (Pixel)	**0.0752**	0.0768	0.0764
LPIPS	0.3719	**0.3619**	0.3676

Patch visualization:

Comparison with Bilinear Upscale (full images):



THANK YOU